

OIS-009 — Decision Engine

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-008
 - OAS-016 — Decision Architecture
 - OCS-000 through OCS-032
-

Purpose

This specification defines the implementation responsibilities of the Decision Engine.

The Constitution defines the nature of Decisions.

The Architecture defines the organization of Decisions.

This specification defines the executable engine that realizes constitutional decision processes while preserving reasoning, alternatives, uncertainty, and historical continuity.

The Decision Engine SHALL execute Decisions.

It SHALL never become the source of constitutional authority.

1. Implementation Objective

The objective of the Decision Engine is to operationalize constitutional decision-making without compromising constitutional understanding.

The Engine SHALL preserve:

Decision identity.

Decision lineage.

Reasoning.

Evidence.

Alternatives.

Uncertainty.

Subsequent learning.

Execution SHALL remain subordinate to constitutional understanding.

2. Decision Lifecycle

Every Decision SHALL support the following operational lifecycle.

Question

↓

Reasoning

↓

Candidate Decisions

↓

Evaluation

↓

Commitment

↓

Execution

↓

Observation

↓

Reflection

↓

Learning ``

Every stage SHALL remain historically reconstructable.

3. Decision Context

The Engine SHALL maintain the constitutional context of every Decision.

Context SHALL include:

Applicable Place.

Applicable Schema.

Relevant Conversations.

Evidence.

Reasoning.

Voices.

Historical decisions.

Current constitutional understanding.

Context SHALL remain explicit.

4. Candidate Decisions

The Engine SHALL preserve multiple candidate Decisions.

Candidate Decisions SHALL remain:

Observable.

Comparable.

Reviewable.

Historically attributable.

Rejected alternatives SHALL remain constitutionally visible.

5. Decision Commitments

Operational commitment SHALL generate immutable constitutional Events.

Commitment SHALL preserve:

Decision identity.

Reasoning state.

Applicable evidence.

Context.

Operational metadata.

Commitment SHALL never erase alternatives.

6. Decision Execution

Execution SHALL remain operationally independent from constitutional validity.

Successful execution SHALL NOT imply constitutional correctness.

Failed execution SHALL NOT invalidate constitutional reasoning.

Execution SHALL become constitutional history.

7. Decision Review

The Engine SHALL support continuous review.

Review SHALL preserve:

Original reasoning.

Observed outcomes.

New evidence.

Updated understanding.

Future revisions.

Review SHALL extend constitutional history.

8. Decision Federation

Decision Engines MAY exchange constitutional Decisions.

Federation SHALL preserve:

Local ownership.

Canonical identity.

Reasoning lineage.

Evidence relationships.

Historical attribution.

Decision federation SHALL remain asynchronous.

9. Recovery

Recovery SHALL reconstruct Decisions using:

Identity.

Event history.

Reasoning.

Evidence.

Architectural relationships.

Recovery SHALL preserve constitutional continuity.

10. Verification

Verification SHALL demonstrate:

Reasoning preservation.

Alternative preservation.

Historical reconstruction.

Execution independence.

Equivalent constitutional decision behavior.

Implementation SHALL remain replaceable.

11. Decision Pattern

The Decision Engine SHALL optimize operational execution.

It SHALL NOT optimize away uncertainty, alternatives, reasoning, or review.

Operational efficiency SHALL remain subordinate to constitutional integrity.

12. Decision Engine Test

A Decision Engine satisfies this specification when independent implementations reconstruct constitutionally equivalent Decisions, including alternatives, reasoning, uncertainty, execution history, and subsequent learning, despite differences in workflow engines, execution models, or infrastructure.

Conformance

An implementation conforms to this specification when Decisions remain operationally executable, constitutionally attributable, historically continuous, architecturally compatible, and independently reconstructable across implementation technologies.

Implementation Principle

Most decision systems optimize for choosing.

Organodynamics optimizes for understanding.

The Decision Engine therefore exists not to produce answers,

but to preserve the complete constitutional journey through which Questions become Decisions,

Decisions become Actions,

and Actions become the beginning of deeper constitutional understanding.

End of Specification OIS-009